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DPP

DAILY PRACTICE PROBLEMS

MATHEMATICS

TARGET: JEE (Main + Advanced) 2022

COURSE : VIKAAS (01JA TO 07JA) **DPPs - A1 to A18**

INDEX

NOTE :

- ✍ **"Marked questions are recommended for Revision"**

DPP No. # A1 (JEE–ADVANCED)

Total Marks : 31		**Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q.1 to Q.7	**(3 marks 3 min.)**	**[21, 21]**
Multiple choice Objective Questions ('–1' negative marking) Q.8	**(4 marks 3 min.)**	**[04, 03]**
Subjective Questions ('–1' negative marking) Q.9 to Q.10	**(3 marks 3 min.)**	**[06, 06]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Information about number system (Q. No. 1 to 7)

Counting numbers have fascinated human mind from time immemorial. The first set he seems to have pondered about is the set of natural numbers, N. Various subsets of this set were defined. Note worthy among them are

<u>Prime Number</u> :- If a natural number has exactly two divisors it is called a prime number. Yet another way to define it is as a natural number, other than 1, which is divisible by 1 & it self only.
Simple examples are 2, 3, 5, 7, ...
{2, 3} in the only set of consecutive primes.

<u>Composite numbers</u> :- A natural number having more than 2 divisors is called a composite number.
Simple examples are 4, 6, 8, 9, 10, ...
Note that 1 is neither prime nor composite.

<u>Coprime or relatively prime numbers</u> :- A pair of natural numbers is called a set of coprime numbers if their highest common factor (HCF) or greatest common divisor (g.c.d.) is 1.
For example 8 & 5 are co-prime
Note that these two numbers need not be prime.
More over 1 is coprime with every natural numbers.
A prime number is coprime with all natural numbers which are not it's multiple.

<u>Twin Prime</u> :- A pair of primes is called twin primes if their non-negative difference is '2'
For example {3, 5}, {5, 7}, {11, 13},..........

The natural numbers were not sufficient to deal with various equations that mathematicians encountered so some new sets of numbers were defined

<u>Whole Numbers (W)</u> = {0, 1, 2, 3, 4, ...}

<u>Integers (Z or I)</u> = {......, –3, –2, – 1, 0, 1, 2, 3, 4, ...}

<u>Even Integers</u> :- Integers divisible by 2, they are expressed as 2n, $n \in Z$.

<u>Odd Integers</u> :- Integers not divisible by 2, they are expressed as 2n + 1 or 2n – 1, $n \in Z$.
Based on above definitions solve the following problems

1. Number of prime numbers less than 10 is p and number of composite numbers less than 15 is q then p + q is equal to
(A) 11 (B) 9 (C) 7 (D) 15

2. Let p & q be the number of natural numbers which are less than or equal 20 and are prime & composite respectively, then 20 – p – q is equal to
(A) 1 (B) 0 (C) 2 (D) 3

3. Difference of squares of two odd integers is always divisible by
(A) 3 (B) 5 (C) 16 (D) 8

4. Identify the correct statement
(A) If a, b, c are odd integers a + b + c cannot be zero
(B) If a, b, c are odd integers $a^2 + b^2 - c^2 \neq 0$
(C) If $a^2 + b^2 = c^2$, then at least one of a, b, c is even, given that a, b, c are integers
(D) If $a^2 + b^2 = c^2$ where a, b, c are integers then c > a + b

5. If $m^2 - n^2 = 7$, where $m, n \in Z$, then number of ordered pairs (m, n) is
Hint : $(m + n)(m - n) = 7 \times 1 = (-7) \times (-1)$
(A) 1 (B) 2 (C) 3 (D) 4

6. Number of ordered pairs of integers (n, m) for which $n^2 - m^2 = 14$ is
(A) 0 (B) 1 (C) 2 (D) 4

7. If $n^2 + 2n - 8$ is a prime number where $n \in N$, then n is
(A) also a prime number (B) relatively prime to 10
(C) relatively prime to 6 (D) a composite number

8. Which of the following collections are sets?
(A) Collection of all natural numbers lying between 21 and 210.
(B) Collection of all rational numbers which lie between $\frac{1}{3}$ and $\frac{1}{2}$.
(C) Collection of handsome boys in class XI of a given school.
(D) Collection of all rectangles in a given plane.

9. Write the following sets in the tabular form :
(i) B = {x : x is a natural number < 10}
(ii) C = {x : x is an odd positive integer and $x^2 < 30$}
(iii) D = {x : x is a letter of the English alphabet in the word 'LALLY'}
(iv) E = {x : x is a natural number that divides 24}.

10. Write the following sets in the set builder form :
(i) P = {2} (ii) Q = {1, 4, 9, 16} (iii) T = {A, E, I, O, U}(iv) V = {0, 3, 6, 9}

DPP No. # A2 (JEE–ADVANCED)

Total Marks : 30 **Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q.1 to Q.7 (3 marks 3 min.) [21, 21]
Subjective Questions ('–1' negative marking) Q.8 to Q.10 (3 marks 3 min.) [09, 09]

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Information about number system (Q. No. 1 to 7)

The number system consisting of integers and its subsets lead to substantial insight in mathematical churning, yet several hurdles were encountered in dealing with plethora of other mathematical equations especially those of polynomial equations. Hence a need was felt to extend the known set of numbers. This paved way for defining rational numbers, irrational numbers and real numbers.

<u>Rational Numbers (Q)</u> :- Numbers which can be expressed in the form $\frac{p}{q}$, $p,q \in I$, $q \neq 0$. Terminating and recurring decimals are also rational numbers.
Note that all integers are also rational numbers

Ex :- $\frac{2}{3}, \frac{4}{9}, \frac{11}{3}, 0.1\overline{23}, 3.1\overline{25}$....

<u>Irrational Numbers</u> :- Real numbers which cannot be expressed in the form $\frac{p}{q}$, $p,q \in I$, $q \neq 0$ are called irrational numbers. Non-terminating and non-recurring decimals are irrational numbers.

Ex :- $\sqrt{2}\ \sqrt[3]{3}$, π, e.

value of π is generally approximated by $\frac{22}{7}$, 3.14, $\frac{355}{113}$.

value of e is generally approximated by 2.71828

Note that $\pi \neq \frac{22}{7}$, $\pi \neq 3.14$, $\pi \neq \frac{355}{113}$, $e \neq 2.71828$

Infact $3.14 < \pi < \frac{22}{7}$.

It is noteworthy that irrational numbers are not defined as what they are instead they are defined as what they are not. Hence if a number is to be proved as an irrational number there is no direct way. We generally assume it to be a rational number which upon further calculation leads to a contradiction, thus establishing the fact that it is an irrational number.

Set of Real numbers (R) is set consisting of rational and irrational numbers.

Given below are some trivial methods of dealing with problems involving rational and irrational numbers.

Note -1 : If a, b, c, d $\in$ Q and λ is an irrational number such that $a + b\lambda = c + d\lambda \Rightarrow a = c$ & $b = d$
In other words we compare rational & irrational terms on both the sides, for example

(i) If b, c $\in$ Q $\quad 2 + b\sqrt{5} = c + 7\sqrt{5} \Rightarrow c = 2$ and $b = 7$

(ii) If a, b $\in$ Q such that $\frac{3+\sqrt{2}}{2+\sqrt{2}} = a + b\sqrt{2}$, then

LHS = $\frac{3+\sqrt{2}}{2+\sqrt{2}} . \frac{2-\sqrt{2}}{2-\sqrt{2}} = \frac{4-\sqrt{2}}{4-2} = \frac{4-\sqrt{2}}{2} = a + b\sqrt{2}$ (RHS) $\Rightarrow$ $a = 2$ and $b = -\frac{1}{2}$

Note-2 : If x is a recurring decimal then it is a rational number and we can always express it as $\frac{p}{q}$, $q \neq 0$, $p, q, \in I$, for example.

(i) $x = 0.\overline{12} \Rightarrow 100x = 12.\overline{12}$

subtracting we get $99x = 12 \Rightarrow x = \frac{12}{99} = \frac{4}{33}$

(ii) $x = 0.2\overline{7} \Rightarrow 10x = 2.\overline{7} \Rightarrow 100x = 27.\overline{7}$

subtracting we get $90x = 25 \Rightarrow x = \frac{25}{90} = \frac{5}{18}$

(iii) $x = 2.1\overline{23} \Rightarrow 10x = 21.\overline{23} \Rightarrow 1000x = 2123.\overline{23}$

subtracting we get $990x = 2102 \Rightarrow x = \frac{2102}{990} = \frac{1051}{445}$

1. Which of the following number is irrational

(A) $\sqrt{\frac{4}{9}}$ (B) $\sqrt[3]{\frac{8}{27}}$ (C) $\frac{7\pi}{22}$ (D) $\frac{22}{7}$

2. If p, q $\in$ N and $0.1\overline{2} = \frac{p}{q}$ where p and q are relatively prime then identify which of the following is incorrect

(A) p is a prime number (B) q – p is a prime number
(C) q + p is a prime number (D) q is a prime number

3. Consider the following statements

(i) The sum of a rational number with an irrational number is always irrational.
(ii) The product of two rational numbers is always rational.
(iii) The product of two irrationals is always irrationals.
(iv) The sum of two rational is always rational.
(v) The sum of two irrationals is always irrational.

The correct order of True/False of above statements is :

(A) T F T F F (B) F F T T T (C) T T F T F (D) T T F F T

4. Let a, b$\in$Q such that $\dfrac{39\sqrt{2}-5}{3-\sqrt{2}} = a + b\sqrt{2}$, then

(A) $\sqrt{\dfrac{b}{a}}$ is a rational number
(B) b and a are coprime rational numbers
(C) b – a is a composite number
(D) $\sqrt{a+b}$ is a rational number

5. Identify the correct statement

(A) If $\sqrt{x} \in Q \Rightarrow x \in Q$
(B) If $x^2 \in Q$ and $x^7 \in Q \Rightarrow x \in Q$
(C) If $x^3 \in Q$ and $x^7 \in Q \Rightarrow x \in Q$
(D) If $x^4 \in Q$ and $x^{11} \in Q \Rightarrow x \in Q$

6. The equation $7x^2 - (7\pi + 22)x + 22\pi = 0$ has

(A) equal roots
(B) a root which is negative
(C) rational roots only
(D) a rational root and an irrational root.

7. There are four prime numbers written in ascending order. The product of the first three is 385 and that of the last three is 1001. The last number is :

(A) 11 (B) 13 (C) 17 (D) 19

8. Find the sum $\dfrac{1}{1+\sqrt{2}} + \dfrac{1}{\sqrt{2}+\sqrt{3}} + \dfrac{1}{\sqrt{3}+\sqrt{4}} + \ldots\ldots\ldots$ upto 99 terms.

9. State, which of the following sets are finite and which infinite :

(i) {x : x is a positive integral root of $x^2 - 2x - 15 = 0$}
(ii) {x : x is a human being in the world}
(iii) {x : x is a multiple of 3}
(iv) {x : x is a real number between 1 and 2}.

10. In each of the following cases, state whether A = B or A $\neq$ B.

(i) A = {a, b, c, d, e}, B = {x : x is first five letters of English alphabet}
(ii) A = {0}, B = {x : x is a natural number less than 1}
(iii) A = {1, 5, 25, 125}, B = {x : x is a positive divisor of 125}
(iv) A = {x : x is a multiple of 10}, B = {10, 15, 20, 25, 30, ...}

DPP No. # A3 (JEE–MAIN)

Total Marks : 30		Max. Time : 30 min.
Comprehension ('–1' negative marking) Q.1 to Q.7	**(3 marks, 3 min.)**	**[21, 21]**
Subjective Questions ('–1' negative marking) Q. 8 to Q.10	**(3 marks, 3 min.)**	**[09, 09]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Information about basic identities (Q. No. 1 to 7)

If a statement is true for all the values of the variable, such statements are called as identities. some basic identities are **:**

(i) $(a + b)^2 = a^2 + 2ab + b^2 = (a - b)^2 + 4ab$
(ii) $a^2 - b^2 = (a + b)(a - b)$
(iii) $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$
(iv) $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = (a + b)(a^2 + b^2 - ab)$
(v) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = a^2 + b^2 + c^2 + 2abc\left(\dfrac{1}{a} + \dfrac{1}{b} + \dfrac{1}{c}\right)$
(vi) $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
$= \dfrac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$
If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

1. If $x + \frac{1}{x} = 2$, then $x^2 + \frac{1}{x^2}$ is equal to

(A) 0 (B) 1 (C) 2 (D) 3

2. If $\left(a+\frac{1}{a}\right)^2 = 3$, then $a^3 + \frac{1}{a^3}$ equals :

(A) 0 (B) $3\sqrt{3}$ (C) $7\sqrt{7}$ (D) $6\sqrt{3}$

3. If $a + b + c = 2$, $a^2 + b^2 + c^2 = 1$ and $abc = 3$ then $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ is equal to

(A) 0 (B) 1/2 (C) 1 (D) 2

4. If $a + b + c = 3x$ then the value of $(x - a)^3 + (x - b)^3 + (x - c)^3 - 3(x - a)(x - b)(x - c) =$

(A) abc (B) 3abc (C) 0 (D) 2abc

5. If $x^3 + y^3 + 1 = 3xy$, where $x \neq y$ determine the value of $x + y + 1$.

6. If $x = \sqrt{3}+\sqrt{2}$, then find $x + \frac{1}{x}$, $x^2 + \frac{1}{x^2}$, $x^3 + \frac{1}{x^3}$, $x^4 + \frac{1}{x^4}$

7. The number of real roots of the equation, $(x - 1)^2 + (x - 2)^2 + (x - 3)^2 = 0$ is :

(A) 0 (B) 1 (C) 2 (D) 3

8. Find the number of positive integers x for which $f(x) = x^3 - 8x^2 + 20x - 13$, is a prime number.

9. Which of the following sets are empty sets?

(i) $\{x : x^2 = 2$ and x is a rational number$\}$

(ii) {x : x is an integer neither positive nor negative}

(iii) {x : x is a boy student in a girls college}

(iv) $\{x : x < 5$ and also $x > 5\}$

(v) $\{x : x$ is a real number and $x^2 < 0\}$

10. Which of the following statements are true and which false?

(i) $0 \in \{\ \}$ (ii) $0 \in \{0\}$ (iii) $0 \in \{\{0\}\}$ (iv) $0 \in \{0, \{0\}\}$

(v) $\{0\} \in \{0\}$

DPP No. # A4 (JEE–ADVANCED)

Total Marks : 30		Max. Time : 30 min.
Comprehension ('–1' negative marking) Q.1 to Q.5	(3 marks, 3 min.)	[15, 15]
Single choice Objective ('–1' negative marking) Q.6 to Q.7	(3 marks, 3 min.)	[06, 06]
Subjective Questions ('–1' negative marking) Q.8 to Q.10	(3 marks, 3 min.)	[09, 09]

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension (Q. No. 1 to 5)

If a function f is defined by f (x) = $a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + ... + a_{n-1} x + a_n$ where n is a non negative integer and $a_0, a_1, a_2, ..., a_n$ are real numbers and $a_0 \neq 0$, then f is called a polynomial function of degree n. For polynomials we can define the following theorem

(i) **Remainder theorem :** Let p(x) be any polynomial of degree greater than or equal to one and 'a' be any real number. If p(x) is divided by (x – a), then the remainder is equal to p(a).

(ii) **Factor theorem :** Let p(x) be a polynomial of degree greater than or equal to 1 and 'a' be a real number such that p(a) = 0, then (x – a) is a factor of p(x). Conversely, if (x – a) is a factor of p(x), then p(a) = 0.

1. The factor of the polynomial $x^3 + 3x^2 + 4x + 12$ is
(A) x + 3 (B) x – 3 (C) x + 2 (D) x – 2

2. The remainder when the polynomial $P(x) = x^4 - 3x^2 + 2x + 1$ is divided by x – 1 is
(A) 0 (B) 1 (C) 2 (D) 3

3. The polynomials $P(x) = kx^3 + 3x^2 - 3$ and $Q(x) = 2x^3 - 5x + k$, when divided by (x – 4) leave the same remainder. Then the value of k is
(A) 2 (B) 1 (C) 0 (D) –1

4. Let f(x) be a polynomial function. If f(x) is divided by x–1, x+1 & x+2, then remainders are 5, 3 and 2 respectively. When f(x) is divided by $x^3 + 2x^2 - x - 2$, then remainder is :
(A) x – 4 (B) x + 4 (C) x – 2 (D) x + 2

5. If (x – a) is a factor of $x^3 - a^2x + x + 2$, then 'a' is equal to
(A) 0 (B) 2 (C) –2 (D) 1

6. Let $N = (2 + 1)(2^2 + 1)(2^4 + 1) .... (2^{32} + 1) + 1$ and $N = 2^\lambda$ then the value of λ is
(A) 63 (B) 64 (C) 65 (D) 66

7. If $(x + y)^2 = 2(x^2 + y^2)$ and $(x - y + \lambda)^2 = 4$, $\lambda > 0$, then λ is equal to :
(A) 1 (B) 2 (C) 3 (D) 4

8. Find the power set of the set {a, b, c}.

9. Let A = {1, 2, 3, 5}, B = {1, 2, 3} and C = {1, 2, 5}. Find all the sets X satisfying.
(i) $X \subset A, X \not\subset B$ (ii) $X \subset A, X \not\subset C$
(iii) $X \subset B, X \neq B, X \not\subset C$ (iv) $X \subset A, X \subset B, X \subset C$

10. Let A = $\{\phi, \{\phi\}, 2, \{2, \phi\}, 3\}$, which of the following are true?
(i) $\phi \in A$ (ii) $\phi \subset A$ (iii) $\{\phi\} \in A$
(iv) $\{\phi\} \subset A$ (v) $2 \subset A$ (vi) $\{2, \phi\} \subset A$
(vii) $\{\{2\}, \{3\}\} \subset A$ (viii) $\{2, 3\} \not\subset A$ (ix) $\{\phi, 2, 3\} \subset A$.

DPP No. # A5 (JEE–ADVANCED)

Total Marks : 31		Max. Time : 30 min.
Comprehension ('–1' negative marking) Q.1 to Q.4	(3 marks, 3 min.)	[12, 12]
Single choice Objective ('–1' negative marking) Q.9	(3 marks, 3 min.)	[03, 03]
Multiple choice objective ('–1' negative marking) Q.10	(4 marks, 3 min.)	[04, 03]
Subjective Questions ('–1' negative marking) Q.5 to Q.8	(3 marks, 3 min.)	[12, 12]

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension (Q.1 to Q.4)

When two ratios are equal, then the four quantities compositing them are said to be proportional.

If $\frac{a}{b} = \frac{c}{d}$, then it is written as a : b = c : d or a : b :: c : d. Also

if $\frac{a}{b} = \frac{c}{d} = \lambda \quad \Rightarrow \quad a = b\lambda$ and $c = d\lambda$

$$\Rightarrow \quad \frac{a}{b} = \frac{c}{d} = \frac{a \pm c}{b \pm d} = \frac{\sqrt[n]{(a)^n \pm (c)^n}}{\sqrt[n]{(b)^n \pm (d)^n}} = \lambda \;;\; (\lambda > 0)$$

important property of proportion :

(i) If a : b = c : d, then

$\frac{a+b}{b} = \frac{c+d}{d}$ (Componendo) i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{b} + 1 = \frac{c}{d} + 1$

(ii) If a : b = c : d, then

$\frac{a-b}{b} = \frac{c-d}{d}$ (Dividendo) i.e. $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a}{b} - 1 = \frac{c}{d} - 1$

(iii) If a : b = c : d, then

$\frac{a+b}{a-b} = \frac{c+d}{c-d}$ (Componendo and dividendo)

1. If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f}$ and $\frac{2a^4b^2 + 3a^2c^2 - 5e^4f}{2b^6 + 3b^2d^2 - 5f^5} = \left(\frac{a}{b}\right)^n$ then the value of n is :

(A) 1 (B) 2 (C) 3 (D) 4

2. If $\frac{a+3d}{a+9d} = \frac{a+d}{a+5d}$ = k, then k is equal to (a, d > 0)

(A) 1/2 (B) 2 (C) 6 (D) 1/4

3. If $\frac{x^3+x^2+x+1}{x^3-x^2+x-1} = \frac{x^2+x+1}{x^2-x+1}$, then the number of real value of x satisfying are

(A) 0 (B) 1 (C) 2 (D) 3

4. The value of x satisfying the equation $\frac{6x+2a+3b+c}{6x+2a-3b-c} = \frac{2x+6a+b+3c}{2x+6a-b-3c}$ is

(A) ab/c (B) 2ab/c (C) ab/3c (D) ab/2c

Solve the following inequations

5. $\frac{x-2}{x^2-9} < 0$

6. $(x + 1)(x - 3)^2(x - 5)(x - 4)^2(x - 2) < 0$

7. $\dfrac{(x-1)\ (x+2)^2}{-1-x} < 0$

8. $\dfrac{1+x^2}{x^2-5x+6} < 0$

9. Least integral value of x satisfying the inequation $(x^2 + 1) < (x + 2)^2 < 2x^2 + 4x - 12$ is

(A) – 2 (B) – 5 (C) 2 (D) 5

10. The solution set of the inequality $\dfrac{1}{x-2} - \dfrac{1}{x} \le \dfrac{2}{x+2}$ is $(-\alpha, \beta] \cup (\gamma, \alpha) \cup [\delta, \infty)$, then

(A) $\alpha + \beta + \gamma + \delta = 5$ (B) $\alpha\beta\gamma\delta = -4$ (C) $\beta\delta = -2$ (D) $\alpha\beta\gamma = 0$

DPP No. # A6 (JEE–MAIN)

Total Marks : 31		**Max. Time : 30 min.**
Single choice Objective ('–1' negative marking) Q.1,2,3,8,9,10	**(3 marks, 3 min.)**	**[18, 18]**
Multiple choice objective ('-2' negative & Partial marking) Q.4	**(4 marks 3 min.)**	**[04, 03]**
Subjective Questions ('–1' negative marking) Q.5 to Q.7	**(3 marks, 3 min.)**	**[09, 09]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. If $2x^3 - 5x^2 + x + 2 = (x - 2)(ax^2 - bx - 1)$, then a & b are respectively :

(A) 2, 1 (B) 2, – 1 (C) 1, 2 (D) – 1, 1/2

2. If x, y are rational numbers such that $(x + y) + (x - 2y)\sqrt{2} = 2x - y + (x - y - 1)\sqrt{5}$ then

(A) x = 1, y = 1 (B) x = 2, y = 1
(C) x = 5, y = 1 (D) x & y can take infinitely many values

3. If log15 = a and log75 = b, then $\log_{75}45$ is :

(A) $\dfrac{3b-a}{a}$ (B) $\dfrac{b-3a}{a}$ (C) $\dfrac{3a-b}{b}$ (D) $\dfrac{a-3b}{b}$

4. $\log_3\left(1+\dfrac{1}{3}\right) + \log_3\left(1+\dfrac{1}{4}\right) + \log_3\left(1+\dfrac{1}{5}\right) + .. + \log_3\left(1+\dfrac{1}{242}\right)$ when simplified has the value equal to :

(A) 2 (B) 4 (C) 6 (D) $\log_2 16$

5. Which of the following sets are equal ?
A = {x : x ∈ N, x < 4}, B = {1, 1, 2, 3, 3}, C = {1, 3}, D = {x : x is an odd natural number < 5}
E = {1, 2, 3}, F = {1, 1, 3}

6. Let U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} and A = {1, 3, 5, 7, 9}. Find A′.

7. Let A = {1, 2, 3, 4, 5, 6}, B = {2, 4, 6, 8}. Find A – B and B – A.

8. If A, B, C are any three sets, then A ∪ (B ∩ C) is

(A) (A ∪ B) ∩ (A ∪ C) (B) (A ∪ B) (A ∪ C) (C) (A ∪ B) ∪ (A ∩ C) (D) None of these

9. A – B is equal to :

(A) B – A (B) A ∪ B (C) A ∩ B (D) A – (A ∩ B)

10. If A and B are any two sets, then (A ∪ B) – (A ∩ B) is equal to

(A) A – B (B) B – A (C) (A – B) ∪ (B – A) (D) None of these

DPP No. # A7 (JEE–ADVANCED)

Total Marks : 31		**Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q. 1 to Q.3	**(3 marks 3 min.)**	**[09, 09]**
Single choice Objective ('–1' negative marking) Q.4 to 9	**(3 marks 3 min.)**	**[18, 18]**
Multiple choice objective ('-2' negative & Partial marking) Q.10	**(4 marks 3 min.)**	**[04, 03]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension (Q. No. 1 to 3)

A number of the form a + ib is called a complex number, where $a,b \in R$ and $i = \sqrt{-1}$. Complex number is usually denoted by Z and the set of complex number is represented by C. Thus $C = \{a + ib : a, b \in R$ and $i = \sqrt{-1}\}$. If Z = a + ib is a complex number then $\bar{z} = a - ib$ is called as conjugate. Two complex numbers $z_1 = a_1 + ib_1$ & $z_2 = a_2 + ib_2$ are equal if and only if their real and imaginary parts are equal respectively. i.e. $z_1 = z_2 \Leftrightarrow Re(z_1) = Re(z_2)$ and $Im(z_1) = Im(z_2) \Leftrightarrow a_1 = a_2$ and $b_1 = b_2$. The following algebric operations can be performed on complex numbers.

1. Addition $(a + bi) + (c + di) = a + bi + c + di = (a + c) + (b + d) i$
2. Subtraction $(a + bi) - (c + di) = a + bi - c - di = (a - c) + (b - d) i$
3. Multiplication $(a + bi) (c + di) = ac + adi + bci + bdi^2 = (ac - bd) + (ad+ bc)i$
4. Division $\dfrac{a+bi}{c+di} = \dfrac{a+bi}{c+di} \cdot \dfrac{c-di}{c-di} = \dfrac{ac+bd}{c^2+d^2} + \dfrac{bc-ad}{c^2+d^2}\, i$

Using above comprehension answer the followings :

1. If $(x^2 + x) + iy$ and $(-x - 1) - i(x + 2y)$ are conjugate of each other, then real value of x & y are
(A) x = – 1, y = 1 (B) x = 1, y = –1 (C) x = 1, y = 1 (D) x = – 1, y = –1

2. $\left(\dfrac{4i^3 - i}{2i+1}\right)^2$ can be expressed in a + ib as
(A) 3 + 4i (B) 3 – 4i (C) 4 + 3i (D) 4 – 3i

3. If $a^2 + b^2 = 1$. Then $\dfrac{1+b+ia}{1+b-ia} =$
(A) 1 (B) 2 (C) b + ia (D) a + ib

4. If $\log_{10}(x - 1)^3 - 3\log_{10}(x - 3) = \log_{10} 8$, then $\log_x 625$ has the value equal to :
(A) 5 (B) 4 (C) 3 (D) 2

5. If $\dfrac{1}{3}\log_3 M + 3\log_3 N = 1 + \log_{0.008}5$, then
(A) $M^9 = \dfrac{9}{N}$ (B) $N^9 = \dfrac{9}{M}$ (C) $M^3 = \dfrac{3}{N}$ (D) $N^9 = \dfrac{3}{M}$

6. If $\log_{10}\dfrac{1025}{1024} = \alpha$ and $\log_{10} 2 = \beta$, then the value of $\log_{10}4100$ in terms of α and β is equal to
(A) $\alpha + 9\beta$ (B) $\alpha + 12\beta$ (C) $12\alpha + \beta$ (D) $9\alpha + \beta$

7. If $x = 2 + 2^{2/3} + 2^{1/3}$, then the value of $(x^3 - 6x^2 + 6x)$ is
(A) 3 (B) 2 (C) 1 (D) None of these

8. If $x = \log_k b = \log_b c = \dfrac{1}{2}\log_c d$, then $\log_k d$ is equal to
(A) 6x (B) $\dfrac{x^3}{2}$ (C) $2x^3$ (D) $2x^8$

9. Let n be an integer greater than 1 and let $a_n = \frac{1}{\log_n 1001}$. If $b = a_3 + a_4 + a_5 + a_6$ and $c = a_{11} + a_{12} + a_{13} + a_{14} + a_{15}$. Then value of (b – c) is equal to

(A) 1001 (B) 1002 (C) – 2 (D) –1

10. If $B = \frac{12}{3+\sqrt{5}+\sqrt{8}}$ and $A = \sqrt{1}+\sqrt{2}+\sqrt{5}-\sqrt{10}$, then value of $\log_A B$ is

(A) a positive integer (B) a prime integer
(C) a non-negative integer (D) non integer

DPP No. # A8 (JEE–ADVANCED)
Special DPP on " Logarithm Table"

Total Marks : 30		**Max. Time : 30 min.**
Single choice Objective ('–1' negative marking) Q.1,2,3	**(3 marks, 3 min.)**	**[09, 09]**
Subjective Questions ('–1' negative marking) Q.4 to Q.10	**(3 marks, 3 min.)**	**[21, 21]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. Number of integers whose characteristic of logarithms to the base 10 is 3, is

(A) 8999 (B) 9000 (C) 90000 (D) 99000

2. If mantissa of lagarithm of 719.3 to the base 10 is 0.8569, then mantissa of logarithm of 71.93 is

(A) 0.8569 (B) $\bar{1}.8569$ (C) 1.8569 (D) 0.1431

3. Number of digits in integral part of $60^{12} + 60^{-12} - 60^{-15}$ is (given log 2 = 0.3030, log 3 = 0.4771)

(A) 20 (B) 21 (C) 22 (D) 24

4. Find logarithm of the following values :

(i) 0.128 (ii) 0.0125 (iii) 36.12 (iv) 0.0002432
(v) 5 (vi) 500 (vii) 0.01361

5. Find antilog of the following values :

(i) $\bar{2}.362$ (ii) – 3.7913 (iii) 2.6329 (iv) 0.0125

6. (i) Find antilog of 0.4 to the base 32. (ii) Find antilog of 2 to the base $\sqrt{3}$.
(iii) Find number whose logarithm is 1.6078.

7. Given $\log_{10}2 = 0.3010$, find $\log_{25} 200$ by using log table

8. Find volume of a cuboid whose edges are 58.73 cm, 2.631 cm and 0.3798 cm using log table.

9. Find the value of $(23.17)^{\frac{1}{5.76}}$ using log table.

10. Find number of digits in 875^{16}

DPP No. # A9 (JEE–ADVANCED)

Total Marks : 35		Max. Time : 30 min.
Comprehension ('–1' negative marking) Q.1 to Q.3	(3 marks, 3 min.)	[09, 09]
Single choice Objective ('–1' negative marking) Q.10	(3 marks, 3 min.)	[03, 03]
Multiple choice objective ('–1' negative marking) Q.5 to Q.9	(4 marks, 3 min.)	[20, 15]
Subjective Questions ('–1' negative marking) Q.4	(3 marks 3 min.)	[03, 03]

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension # 1 (Q. No. 1 to 3)

Given that $N = 7^{\log_{49} 900}$, $A = 2^{\log_2 4} + 3^{\log_2 4} + 4^{\log_2 2} - 4^{\log_2 3}$, $D = (\log_5 49)(\log_7 125)$
Then answer the following questions : (Using the values of N, A, D)

1. If $\log_A D = a$, then the value of $\log_6 12$ is (in terms of a)

(A) $\frac{1+3a}{3a}$ (B) $\frac{1+2a}{3a}$ (C) $\frac{1+2a}{2a}$ (D) $\frac{1+3a}{2a}$

2. If the value obtained in previous question is $\frac{1+ma}{na}$, then choose the correct option

(A) $\log_N m < \log_m N = \log_n N$ (B) $\log_N m < \log_n N < \log_m N$
(C) $\log_m N < \log_N m < \log_n N$ (D) $\log_m N < \log_N m = \log_n N$

3. The value of $\log_{\left(A-\frac{N}{10}\right)} |N + A + D + m + n| - \log_5 2$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

4. Solve $\frac{(x-5)^2 \quad (x+2)^3 \quad (x-4)}{(x-3)^4} \le 0$

5. If x & y are real numbers and $\frac{y}{x}$ = x, then ' y ' cannot take the value(s) :

(A) – 1 (B) 0 (C) 1 (D) 2

6. If $\log_4 5 = x$ and $\log_5 6 = y$ then

(A) $\log_4 6 = xy$ (B) $\log_6 4 = xy$ (C) $\log_3 2 = \frac{1}{2xy-1}$ (D) $\log_2 3 = \frac{1}{2xy-1}$

7. Which is/are true ?

(A) $\log_{0.2} 3 < \log_2 3$ (B) $\log_{\sqrt{5}} 10 < \log_{\sqrt{3}} 11$
(C) $\log_{\sqrt{3}} 10 > \log_{\sqrt{5}} 11$ (D) $\log_{(2-\sqrt{3})}(7-4\sqrt{3}) > \log_{(\sqrt{3}-1)}(4-2\sqrt{3})$

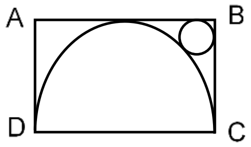
8. P(x) is a polynomial with integral cofficient such that for four distinct integers a, b, c, d; P(a) = P(b) = P(c) = P(d) = 3.If P(e) = 5 (e is an integer), then e can't be

(A) 1 (B) 3 (C) 4 (D) 5

9. The value of $\frac{6a^{\ell nb} \log_{a^2} b \log_{b^2} a}{e^{(\ell na)(\ell nb)}}$ is

(A) independent of a (B) independent of b (C) dependent on a (D) dependent of b

10. The figure shows a rectangle ABCD with a semi-circle and a circle inscribed inside it as shown. What is the ratio of the area of the circle to that of the semi-circle?



(A) $(\sqrt{2}-1)^2$ (B) $2(\sqrt{2}-1)^2$ (C) $(\sqrt{2}-1)^2/2$ (D) None of these

DPP No. # A10 (JEE–ADVANCED)

Total Marks : 31			**Max. Time : 30 min.**
Single choice Objective ('-1' negative marking) Q. 1 to Q.9		**(3 marks 3 min.)**	**[27, 27]**
Multiple choice objective ('-2' negative & Partial marking) Q.10		**(4 marks 3 min.)**	**[04, 03]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. If product of the roots of the equation $3x^2 - 4x + (\log a^2 - \log(-a) + 3) = 0$ is 1, then 'a' is equal to
(A) not possible (B) – 1 (C) 1 (D) None of these

2. The quadratic equation $x^2 - 9x + 3 = 0$ has roots α and β. If $x^2 - bx - c = 0$ has roots α^2 and β^2, then (b, c) is
(A) (75, –9) (B) (–75, 9) (C) (– 87, 4) (D) (–87, 9)

3. If sum of the roots of the quadratic equation, $a x^2 + b x + c = 0$ is 12, then the sum of the roots of the equation, $a (x + 1)^2 + b (x + 1) + c = 0$ is :
(A) 9 (B) 10 (C) 12 (D) 14

4. If a, b, c are real numbers satisfying the condition $a + b + c = 0$ then the roots of the quadratic equation $3ax^2 + 5bx + 7c = 0$ are :
(A) positive (B) negative (C) real & distinct (D) imaginary

5. The statement which gives the best description of location of the vertex of the quadratic trinomial $f(x) = ax^2 + bx + c$, with $a > 0$ and $b^2 - 4ac > 0$, is
(A) It will lie on the y-axis (B) It will lie below the x-axis, on the y-axis
(C) It will lie above the x-axis, on the y-axis (D) It will not lie on x-axis

6. If $a + b + c = 0$ and $a^2 + b^2 + c^2 = 1$, then the value of $a^4 + b^4 + c^4$ is
(A) 1 (B) 4 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

7. If A & B are two rational numbers and AB, A + B and A – B are rational numbers, then A/B is :
(A) always rational (B) never rational
(C) rational when $B \neq 0$ (D) rational when $A \neq 0$

8. If $\ell n^2 x + 3\,\ell n\, x - 4$ is non negative, then x must lie in the interval :
(A) $[e, \infty)$ (B) $(-\infty, e^{-4}) \cup [e, \infty)$ (C) $(1/e, e)$ (D) None of these

9. Complete solution set of inequality $\frac{(\pi^x - 7^x)\log_{10}(x-4)}{(x^2 - 9x + 18)(x^2 - x)} < 0$ is
(A) $x \in R - \{0,1,3,6\}$ (B) $x \in (4, 5) \cup (6, \infty)$
(C) $x \in (5, 6)$ (D) $x \in (4, \infty)$

10. If α and β ($\alpha < \beta$) are values of a for which equation $x^2 - x(1-a) - (a+2) = 0$ has integral roots, then
(A) $|\alpha - \beta| = 2$ (B) $\alpha^2 + \beta^2 = 9$ (C) $\alpha^2 + \beta^2 = 4$ (D) $\alpha^2 - \beta^2 = 4$

DPP No. # A11 (JEE–MAIN)

Total Marks : 30 **Max. Time : 30 min.**
Single choice Objective ('-1' negative marking) Q. 1 to Q.10 **(3 marks 3 min.)** **[30, 30]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. If $-3 + 5i$ is a root of the equation $x^2 + px + q = 0$, then the ordered pair (p, q) is $(p, q \in R)$
(A) $(-6, 34)$ (B) $(6, 34)$ (C) $(34, -6)$ (D) $(34, 6)$

2. If α, β be the roots of the equation $(x - a)(x - b) + c = 0$ $(c \neq 0)$, then the roots of the equation $(x - c - \alpha)(x - c - \beta) = c$ are
(A) a and b + c (B) a + b and b (C) a + c and b + c (D) a – c and b – c

3. Number of non-negative integral values of 'k' for which roots of the equation $x^2 + 6x + k = 0$ are rational is-
(A) 1 (B) 2 (C) 3 (D) 4

4. If $P(x) = ax^2 + bx + c$ and $Q(x) = -ax^2 + dx + c$, $ac \neq 0$, then the equation $P(x) . Q(x) = 0$ has
(A) Exactly two real roots (B) Atleast two real roots
(C) Exactly four real roots (D) No real roots

5. If x_1, x_2 & x_3 are three distinct numbers which satisfy the relation $(a - 1)x^2 + (a^2 - 4a + 3)x + (a^2 + a - 2) = 0$ then
(A) $a = -1$ (B) $a = -2$ (C) $a = 3$ (D) None of these

6. If α, β are the roots of $x^2 - a(x - 1) + b = 0$, then the value of $\frac{1}{\alpha^2 - a\alpha} + \frac{1}{\beta^2 - a\beta} + \frac{2}{a+b}$
(A) $\frac{4}{a+b}$ (B) $\frac{1}{a+b}$ (C) 0 (D) –1

7. Find the value of 'k' for which the following set of quadratic equations has exactly one common root, $x^2 - kx + 10 = 0$ and $x^2 + kx - 18 = 0$.
(A) ± 3 (B) ± 7 (C) ± 9 (D) ± 11

8. a, b, c are reals such that $a + b + c = 3$ and $\frac{1}{a+b} + \frac{1}{b+c} + \frac{1}{c+a} = \frac{10}{3}$.
The value of $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}$ is
(A) 9 (B) 7 (C) 5 (D) 3

9. The multiplication of a rational number 'x' and an irrational number 'y' is :
(A) always rational (B) rational except when $y = \pi$
(C) always irrational (D) irrational except when $x = 0$

10. The number of positive integral solutions of , $\frac{x^2(4-3x)^3(x-2)^4}{(x-5)^5.(2x-7)^6} \geq 0$ is :
(A) 1 (B) 2 (C) 3 (D) 4

DPP No. # A12 (JEE–ADVANCED)

Total Marks : 33		**Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q. 1 to Q.2	**(3 marks 3 min.)**	**[06, 06]**
Single choice Objective ('-1' negative marking) Q. 3	**(3 marks 3 min.)**	**[03, 03]**
Multiple choice objective ('-2' negative & Partial marking) Q.4 to Q.5,8	**(4 marks 3 min.)**	**[12, 09]**
Subjective Questions ('–1' negative marking) Q.6 to Q.7, 9,10	**(3 marks 3 min.)**	**[12, 12]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension # 1 (Q.1 to Q.2)

If $\frac{u}{\alpha} = \frac{v}{\beta}$ then $\frac{u}{\alpha} = \frac{v}{\beta} = \left(\frac{au^n + bv^n}{a\alpha^n + b\beta^n}\right)^{\frac{1}{n}}$

1. If $\frac{a}{b} = \frac{a-c}{b-d}$, then $\left(\frac{2a^2+3c^2}{2b^2+3d^2}\right)^{\frac{1}{2}} =$

(A) $\frac{b}{a}$ (B) $\frac{a-b}{c-d}$ (C) $\frac{c}{d}$ (D) $\frac{c-d}{c+d}$

2. If ratio of the work done by n men in (n + 2) days is to the work done by n + 3 men in n days is $(n^2 + 2n + 9) : (n^2 + 3n + 10)$, then n is equal to

(A) 2 (B) 3 (C) 5 (D) 7

3. If complete solution set of the inequality $(x^2 + x - 2)(x^2 + x - 16) + 40 \geq 0$ is $x \in (-\infty, -4] \cup [a, b] \cup [c, \infty)$ then value of a + b + c is equal to

(A) 0 (B) 1 (C) 2 (D) 3

4. Equation $(x + 2)(x + 3)(x + 8)(x + 12) = 4x^2$ has

(A) four real & distinct roots (B) two irrational roots
(C) two integer roots (D) two imaginary roots

5. If the ratio of the roots of the equation , $a x^2 + b x + c = 0$ is λ, then $\frac{(\lambda+1)^2}{\lambda}$ is equal to

(A) $\frac{b}{a\ c}$ (B) $\frac{b^2}{a\ c}$ (C) $\frac{a\ c}{b^2}$ (D) $\frac{a\ c}{b}$

6. If α, β, γ are roots of $2x^3 + x^2 + 2x - 1 = 0$, then find the value of $\frac{\alpha-1}{\alpha+1} + \frac{\beta-1}{\beta+1} + \frac{\gamma-1}{\gamma+1}$.

7. Find integral values of x for which $x^2 + 9x + 1$ is a perfect square of an integer.

8. If the remainder R(x) = ax + b is obtained by dividing the polynomial x^{100} by the polynomial $x^2 - 3x + 2$ then

(A) $a = 2^{100} - 1$ (B) $b = 2(2^{99} - 1)$ (C) $b = -2(2^{99} - 1)$ (D) $a = 2^{100}$

9. Solve : $\log_{1/4}\left(\frac{35-x^2}{x}\right) \geq -\frac{1}{2}$

10. Solve : $(0.5)^{\log_{1/3}\frac{x+5}{x^2+3}} > 1$

DPP No. # A13 (JEE–ADVANCED)

Total Marks : 32		**Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q. 1 to Q.2	**(3 marks 3 min.)**	**[06, 06]**
Single choice Objective ('-1' negative marking) Q. 3 to Q.7	**(3 marks 3 min.)**	**[15, 15]**
Multiple choice objective ('-2' negative & Partial marking) Q.8, 10	**(4 marks 3 min.)**	**[08, 06]**
Subjective Questions ('–1' negative marking) Q.9	**(3 marks 3 min.)**	**[03, 03]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension # 1 (Q. No. 1 to 2)

Let $f(x) = ax^2 + bx + c$ where $a,b,c \in R$ and $a \neq 0$. It is given that $f(5) = -3f(2)$ and 3 is a root of $f(x) = 0$.

1. The other root of $f(x) = 0$ is
(A) – 4 (B) 2 (C) 6 (D) cannot be determined

2. The value of $(a + b + c)$ is
(A) 9 (B) 14
(C) 13 (D) cannot be uniquely determined

3. The smallest value of $x^2 - 3x + 3$ in the interval $\left[-3, \frac{3}{2}\right]$ is
(A) – 20 (B) – 15 (C) 5 (D) 3/4

4. Given that $ax^2 + bx + c = 0$ has no real roots and $a + b + c < 0$, then –
(A) $c \neq 0$ (B) $c < 0$ (C) $c > 0$ (D) $c = 0$

5. The least integral value of 'm' for which the expression $m x^2 - 4x + 3m + 1$ is positive for every $x \in R$ is :
(A) 1 (B) – 2 (C) – 1 (D) 2

6.✍ The value of 'a' for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - a - 1 = 0$ assumes the least value is
(A) 3 (B) 2 (C) 1 (D) 0.

7. The cubic polynomial P(x) satisfies the condition that $(x - 1)^2$ is a factor of $P(x) + 2$, and $(x + 1)^2$ is a factor of $P(x) - 2$.Then P(3) equals.
(A) 27 (B) 18 (C) 12 (D) 6

8. If quadratic equation $x^2 + 2(a + 2b)x + (2a + b - 1) = 0$ has unequal real roots for all $b \in R$ then the possible values of a can be equal to
(A) 5 (B) – 1 (C) – 10 (D) 3

9. If $x \in R$, then range of $\frac{x-2}{x^2 - x + 2}$ is $[\lambda, \mu]$. Find the value of $10|\lambda| + 7|\mu|$.

10. If $\log_x (2 + x) \leq \log_x (6 - x)$ then x can be
(A) (1, 2] (B) $(0, 1) \cup (1, 2]$ (C) $(0, 1) \cup [2, 6)$ (D) (3/2, 2]

DPP No. # A14 (JEE–MAIN)

Total Marks : 30 **Max. Time : 30 min.**
Single choice Objective ('-1' negative marking) Q. 1 to Q.10 **(3 marks 3 min.)** **[30, 30]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. If a, b, c, d, e, f are in A.P. then e – c is equal to
(A) 2(c – a) (B) 2(f – d) (C) 2(d – c) (D) d – c

2. The sum of integers from 1 to 100 that are divisible by 2 or 5 is
(A) 2550 (B) 1050 (C) 3050 (D) None of these

3. If A and B are any two sets, then -
(A) $(A-B)\cap B = A$ (B) $(A-B)\cap B = A\cup B$ (C) $(A-B)\cap B = \phi$ (D) $(A-B)\cap B = B$

4. In an examination of a certain class, at least 70% of the students failed in Physics, at least 72% failed in Chemistry, at least 80% failed in Mathematics and at least 85% failed in English. How many at least must have failed in all the four subjects ?
(A) 8% (B) 7% (C) 6% (D) 15%
(E) Cannot be determined due to insufficient data

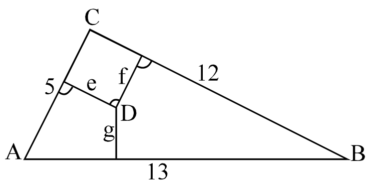
5. Let $y = \cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{2+\cfrac{1}{3+\ldots\ldots\ldots}}}}$, then the value of y is
(A) $\frac{\sqrt{13}+3}{2}$ (B) $\frac{\sqrt{13}-3}{2}$ (C) $\frac{\sqrt{15}+3}{2}$ (D) $\frac{\sqrt{15}-3}{2}$

6. If the roots of the equation $x^2 + 2cx + ab = 0$ are real and unequal, then the roots of the equation $x^2 - 2(a+b)x + (a^2 + b^2 + 2c^2) = 0$ are
(A) real and unequal (B) real and equal (C) imaginary (D) rational

7. If α, β, γ are the roots of the equation $x^3 - px^2 + qx - r = 0$, then the value of $\sum \alpha^2\beta$ is equal to
(A) pq + 3r (B) pq + r (C) pq – 3r (D) q^2/r

8. If $\log_y x = (a \log_z y) = (b \log_x z) = ab$, then which of the following pairs of values for (a, b) is not possible?
(A) (1, 1) (B) (2, 2) (C) (–2, 1/2) (D) (0.4, 2.5)

9. The sides of a triangle ABC are as shown in the given figure. Let D be any internal point of this triangle and let e, f, and g denote the distance between the point D and the sides of the triangle. The sum (5e + 12f + 13g) is equal to



(A) 120 (B) 90 (C) 60 (D) 30

10. When a polynomial P(x) is divided by x, (x – 2) and (x – 3), remainders are 1, 3 and 2 respectively. If the same polynomial is divided by x(x – 2)(x – 3), the remainder is g(x), then the value of g(1) is
(A) 1 (B) 5/3 (C) 2/3 (D) 8/3

DPP No. # A15 (JEE–ADVANCED)

Total Marks : 32		**Max. Time : 30 min.**
Single choice Objective ('-1' negative marking) Q. 1 to Q.6,9	**(3 marks 3 min.)**	**[21, 21]**
Multiple choice objective ('-2' negative & Partial marking) Q.7 to Q.8	**(4 marks 3 min.)**	**[08, 06]**
Subjective Questions ('–1' negative marking) Q.10	**(3 marks 3 min.)**	**[03, 03]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. If both roots of $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3, then :

(A) $a < 2$ (B) $2 \le a \le 3$ (C) $3 < a \le 4$ (D) $a > 4$

2. Let $N = \dfrac{4^5 + 4^5 + 4^5 + 4^5}{3^5 + 3^5 + 3^5} \cdot \dfrac{6^5 + 6^5 + 6^5 + 6^5 + 6^5 + 6^5}{2^5 + 2^5}$ then the value of $\log_2 N$ =

(A) 10 (B) 11 (C) 12 (D) 14

3. If $P = 3^{\sqrt{\log_3 2}} - 2^{\sqrt{\log_2 3}}$ and $Q = \log_2 \log_3 \log_2 \log_{\sqrt{3}} 81$, then

(A) P = 1 , Q = 0 (B) P = 0 , Q = 1 (C) P = Q = 0 (D) P = Q = 1

4. The sum of first 100 terms common to the series 17, 21, 25,.... & 16, 21, 26.... is

(A) 101100 (B) 111000 (C) 110010 (D) 100101

5. Sum of an infinitely many terms of a G.P. is 3 times the sum of even terms. The common ratio of the G.P. is–

(A) 1/2 (B) 2 (C) 3/2 (D) None of these

6.✍ If the sum of the first 2n terms of the A.P. 2, 5, 8, ..., is equal to the sum of the first n terms of the A.P. 57, 59, 61, ..., then n equals

(A) 10 (B) 12 (C) 11 (D) 13

7. If α, β, γ are roots of the equation $x^3 + 3x + 1 = 0$, then $\alpha^4 + \beta^4 + \gamma^4$ = ______

8. If $x^3 + 2x^2 - x + \lambda = 0$ and $x^2 + 5x - 3\lambda = 0$ has a common root, then λ can be

(A) 0 (B) –2 (C) $-\dfrac{14}{27}$ (D) 1

9. $e^{2x} - (a - 1)e^x - a \ge 0 \ \forall \ x \in R$, then values of a will satisfy.

(A) $a \in (-\infty, 0)$ (B) $a \in (-\infty, 0]$ (C) $a \in R$ (D) $a \in (-\infty, -2]$

10. The product of uncommon real roots of the two polynomials $P(x) = x^4 + 2x^3 - 8x^2 - 6x + 15$ and $Q(x) = x^3 + 4x^2 - x - 10$, is

DPP No. # A16 (JEE–ADVANCED)

Total Marks : 31		**Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q. 1 to Q.2	**(3 marks 3 min.)**	**[06, 06]**
Single choice Objective ('-1' negative marking) Q. 3 to Q.8,10	**(3 marks 3 min.)**	**[21, 21]**
Multiple choice objective ('-2' negative & Partial marking) Q.9	**(4 marks 3 min.)**	**[04, 03]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension # 1 (Q. No. 1 to 2)

Consider the quadratic polynomial f(x) = $\frac{x^2}{4} - ax + a^2 + a - 2$, then

1. If the origin lies between zero's of polynomial, then number of integral value(s) of 'a' is
(A) 1 (B) 2 (C) 3 (D) more than 3

2. If roots of equation f(x) = 0 on the number line symmetrically placed about the point 1 then value of a is
(A) $\frac{1}{2}$ (B) $\frac{-1}{2}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

3. If the equation $ax^2 + 2bx - 3c = 0$ has no real roots and $\left(\frac{3c}{4}\right) < a + b$, then –
(A) $c < 0$ (B) $c > 0$ (C) $c = 0$ (D) $a + 2b - 3c > 0$

4. The set of all values of 'a' for which the quadratic equation
$3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possess roots of opposite sign, is
(A) $(-\infty, 1)$ (B) $(-\infty, 0)$ (C) $(1, 2)$ (D) $(3/2, 2)$

5_. If 1 + 6 + 11 + ... + x = 148, then x is equal to
(A) 36 (B) 8 (C) 30 (D) None of these

6. If the sum of first three terms of a G.P. is to the sum of first six terms as 125 : 152, then the common ratio of the G.P. is
(A) $\frac{3}{5}$ (B) $\frac{5}{3}$ (C) $\frac{2}{5}$ (D) $\frac{5}{2}$

7. The ratio of sums of n – terms of two arithmetic progressions is (3 n – 13) : (5 n + 21). The ratio of 24^{th} term of the two series is :
(A) 59 : 141 (B) 7 : 17 (C) 1 : 2 (D) None of these

8. The equation $\log_2(2x^2) + \log_2 x.\ x^{\log_x(\log_2 x+1)} + 1/2\log_4^2(x^4) + 2^{-3\log_{1/2}(\log_2 x)} = 1$ has :
(A) exactly one real solution (B) two real solutions
(C) three real solutions (D) No solution

9. If the equation $\left(\frac{x}{1+x^2}\right)^2 + a\left(\frac{x}{1+x^2}\right) + 3 = 0$ has exactly two real roots which are distinct, then the set of possible real values of a is
(A) $\left(\frac{-13}{2}, 0\right)$ (B) $\left(-\infty, -\frac{13}{2}\right)$ (C) $\left(\frac{-13}{2}, \frac{13}{2}\right)$ (D) $\left(\frac{13}{2}, \infty\right)$

10. The number of positive integers 'n' such that n + 9, 16n + 9, 27n + 9 are all perfect squares is :
(A) 1 (B) 2 (C) 3 (D) 4

DPP No. # A17 (JEE–MAIN)

Total Marks : 30			**Max. Time : 30 min.**
Single choice Objective ('-1' negative marking) Q. 1 to Q.8		**(3 marks 3 min.)**	**[24, 24]**
Subjective Questions ('–1' negative marking) Q.9 to Q.10		**(3 marks 3 min.)**	**[06, 06]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

1. The sum of first p-terms of a sequence is p(p + 1) (p + 2). The 10^{th} term of the sequence is
(A) 396 (B) 600 (C) 330 (D) 114

2. The eighth term of G.P. is 128 and common ratio is 2. The product of its first five terms is
(A) 4^6 (B) 4^3 (C) 4^5 (D) 4^4

3. If $x = \sqrt[3]{7+5\sqrt{2}} - \frac{1}{\sqrt[3]{7+5\sqrt{2}}}$, then the value of $x^3 + 3x - 14$ is equal to
(A) 1 (B) 0 (C) 2 (D) 4

4. The value of the expression , $\log_4\left(\frac{x^2}{4}\right) - 2\log_4(4x^4)$ when $x = -2$ is :
(A) – 6 (B) – 5 (C) – 4 (D) meaningless

5 The solution set of inequality $\frac{(x-5)^{2005}.\ (x+8)^{2008}\ (1-x)}{x^{2006}(x-2)^3\ .\ (x-3)^5\ .\ (x-6)\ (x+9)^{2010}} \geq 0$ is
(A) $(-\infty, -9) \cup (-8, 0) \cup (0, 1) \cup (2, 3) \cup [5, 6)$
(B) $(-\infty, -9) \cup (-9, 0) \cup (0, 1) \cup (2, 3) \cup (5, 6)$
(C) $(-\infty, -9) \cup (-9, 0) \cup (0, 1] \cup (2, 3) \cup [5, 6)$
(D) $(-\infty, 0) \cup (0, 1] \cup (2, 3) \cup [5, 6)$

6 The value of $4\left(3\log_2\frac{81}{80} + 5\log_2\frac{25}{24} + 7\log_2\frac{16}{15}\right)$ is
(A) 1 (B) 2 (C) 4 (D) 5

7 **STATEMENT -1 :** $ln1 + ln2 + ln3 = ln(1 + 2 + 3)$.
STATEMENT -2 : $ln(a + b + c) = lna + lnb + lnc$, where a, b, c are positive.
(A) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1
(B) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
(C) Statement -1 is True, Statement -2 is False
(D) Statement -1 is False, Statement -2 is True

8 If $(x - 1)^3 + (y - 2)^3 + (z - 3)^3 = 3(x - 1)(y - 2)(z - 3)$ and $x - 1 \neq y - 2 \neq z - 3$ then x + y + z is equal to
(A) 2 (B) – 6 (C) 6 (D) 4

9. Find the equation each of whose roots is greater by unity , than the roots of the equation
$x^3 - 5x^2 + 6x - 3 = 0$.

10. Solve the equation $(\log_{10}x)^2 - (\log_{10}x) - 6 = 0$

DPP No. # A18 (JEE–ADVANCED)

Total Marks : 33		**Max. Time : 30 min.**
Comprehension ('–1' negative marking) Q.1 to Q.2	**(3 marks 3 min.)**	**[06, 06]**
Single choice Objective ('-1' negative marking) Q.3 to Q.5	**(3 marks 3 min.)**	**[09, 09]**
Multiple choice objective ('-2' negative & Partial marking) Q.6 to Q.8	**(4 marks 3 min.)**	**[12, 09]**
Subjective Questions ('–1' negative marking) Q.9 to Q.10	**(3 marks 3 min.)**	**[06, 06]**

Question No.	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											

Comprehension# 1 (Q.1 to 2)

Consider the equations

$(3x)^{\log 3} = (4y)^{\log 4}$ and $4^{\log x} = 3^{\log y}$

Answer the following questions

1. Value of x is

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) 3 (D) 4

2. Value of y is

(A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) 3 (D) 4

3. The n^{th} term of the sequence 5 + 55 + 555 + ... is

(A) $5 \times 10^{n-1}$ (B) $5 \times 11^{n-1}$ (C) $\frac{5}{9}(10^n - 1)$ (D) None of these

4. Find the sum of the series $\frac{1}{2} - \frac{1}{2\times 2^2} + \frac{1}{3\times 2^3} - \frac{1}{4\times 2^4} + ...$

Information : $\log_e(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + ...$ to ∞

(A) $\log_e\left(\frac{3}{2}\right)$ (B) $\log_e\left(\frac{2}{3}\right)$ (C) $\log_e\left(\frac{4}{7}\right)$ (D) None of these

5. The largest real solution of the equation $\sqrt{\sqrt{\sqrt{x}}} = \sqrt[4]{\sqrt[4]{\sqrt[4]{3x^4+4}}}$ lies in the interval

(A) $\left(0, \frac{\pi}{6}\right)$ (B) $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ (C) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (D) $\left(\frac{\pi}{2}, \pi\right)$

6. The number N = $\frac{1+2\log_3 2}{(1+\log_3 2)^2} + \log_6^2 2$ when simplified reduces to :

(A) a prime number (B) an irrational number
(C) a real which is less than $\log_3 \pi$ (D) a real which is greater than $\log_7 6$

7. The value of x satisfying

$2\log_{\frac{1}{4}}(x+5) > \frac{9}{4}\log_{\frac{1}{3\sqrt{3}}}(9) + \log_{\sqrt{x+5}}(2)$ is/are

(A) (–5, –4) (B) (–3, –1) (C) (–4, –1) (D) (–5, –2)

8. If p, q $\in$ N satisfy the equation $x^{\sqrt{x}} = (\sqrt{x})^x$ then p & q are

(A) relatively prime
(B) twin prime
(C) Coprime
(D) If $\log_q p$ is defined then $\log_p q$ is not defined vice versa

9. If α, β are the roots of the equation, $x^2 - 2x + 3 = 0$, find the equation whose roots are, $\alpha^3 - 3\alpha^2 + 5\alpha - 2$ and $\beta^3 - \beta^2 + \beta + 5$.

10. Inequality $\frac{ax^2 + 3x + 4}{x^2 + 2x + 2} < 5$ is satisfied for all real values of x then, find out greatest integral value of